

Ubuntu / WSL on Windows

Quick Troubleshooting Guide

For students: find the error you see, then follow the fix.

WINDOWS POWERSHELL

Use for commands beginning with wsl

```
wsl --status
```

UBUNTU TERMINAL

Use for Linux commands

```
sudo apt update
```

1. Installation and startup problems

Error / symptom	What it usually means	Fix
'wsl' is not recognized / WSL is not installed	The Windows WSL feature has not been installed.	PowerShell - Run as Administrator <pre>wsl --install</pre> Restart Windows when prompted, then open Ubuntu.
0x80370102 / virtual machine could not be started	Hardware virtualization or Virtual Machine Platform is disabled.	Windows + BIOS/UEFI 1. Enable Virtual Machine Platform in Windows Features. 2. If it still fails, enable Intel VT-x / Virtualization Technology or AMD SVM / AMD-V in BIOS/UEFI. 3. Restart Windows and try Ubuntu again.
Installing Ubuntu stays at 0%	The normal distribution download is stuck.	PowerShell - Run as Administrator <pre>wsl --install --web-download -d Ubuntu</pre>
WSL works but Ubuntu is missing	WSL is installed, but no Ubuntu distribution is installed.	PowerShell <pre>wsl --list --online</pre> <pre>wsl --install -d Ubuntu</pre>

Error / symptom	What it usually means	Fix
Ubuntu is VERSION 1	The distribution is using WSL 1 rather than WSL 2.	PowerShell <code>wsl -l -v</code> <code>wsl --set-version Ubuntu 2</code>
Ubuntu freezes, will not launch, or networking suddenly stops	The WSL virtual machine may need to be restarted.	PowerShell <code>wsl --shutdown</code> Then reopen Ubuntu.
WSL commands/options are missing or outdated	The installed WSL package is old.	PowerShell <code>wsl --update</code> <code>wsl --shutdown</code> Reopen Ubuntu afterward.

2. Ubuntu, networking, and password problems

Error / symptom	What it usually means	Fix
'Temporary failure resolving...' / apt cannot reach Ubuntu servers	WSL networking or DNS is having a problem; VPNs and institutional networks can contribute.	First try PowerShell <code>wsl --shutdown</code> Reopen Ubuntu and try again. If it still fails, temporarily disconnect a VPN and retry, or restart Windows/network connection.
Nothing appears while typing a sudo password	Linux intentionally hides password characters. This is normal.	Ubuntu Type the password normally and press Enter. No dots or asterisks will appear.
sudo says the password is incorrect	Ubuntu has its own Linux user password; it may not be the Windows password.	Ubuntu / WSL Use the password created the first time Ubuntu was opened. If it was forgotten, the Linux password must be reset.

Error / symptom	What it usually means	Fix
'systemctl' says system has not been booted with systemd	The WSL installation may be old or systemd may not be enabled.	PowerShell first wsl --update wsl --shutdown Reopen Ubuntu. On older distributions, systemd may need to be enabled in /etc/wsl.conf.

3. Files, paths, and scripts

Error / symptom	What it usually means	Fix
cd C:\Users\... does not work in Ubuntu	Ubuntu uses Linux-style paths, not Windows paths.	Ubuntu Convert the Windows path: C:\Users\student\Desktop becomes /mnt/c/Users/student/Desktop
cd fails when the folder name contains spaces	Spaces split command arguments unless the path is quoted.	Ubuntu cd "/mnt/c/Users/student/My Documents"
Permission denied when running ./script.sh	The script may not be executable.	Ubuntu chmod +x script.sh ./script.sh
/bin/bash^M: bad interpreter / '^M' appears	The file was saved with Windows CRLF line endings instead of Linux LF.	Ubuntu sed -i 's/\r\$//' script.sh Also configure the text editor to save shell scripts with LF line endings.
Linux tools are unusually slow in /mnt/c/...	The project is stored on the Windows filesystem. Linux-heavy I/O is often faster in the WSL filesystem.	Ubuntu mkdir -p ~/materials_by_design cd ~/materials_by_design Keep active Linux projects here when possible.

Error / symptom	What it usually means	Fix
Cannot find Ubuntu files in Windows Explorer	The files are in the WSL Linux filesystem.	Ubuntu explorer.exe . This opens the current Ubuntu folder in Windows Explorer.

4. Python and Conda problems

Error / symptom	What it usually means	Fix
'conda: command not found' after installing Conda	The shell has not loaded Conda initialization, or the terminal needs to be restarted.	Ubuntu source ~/.bashrc If that does not work, close and reopen Ubuntu. Confirm the Conda installation completed successfully.
Package installed, but Python says ModuleNotFoundError	pip/conda may have installed the package into a different Python environment.	Ubuntu which python python --version which pip conda env list Activate the intended environment and install packages there. Prefer: python -m pip install PACKAGE
Wrong Conda environment is being used	The expected environment is not active.	Ubuntu conda env list conda activate ENVIRONMENT_NAME which python
PyTorch says CUDA is unavailable	WSL 2, the Windows GPU driver, or the PyTorch CUDA installation may not support GPU access.	PowerShell + Ubuntu wsl -l -v Confirm Ubuntu is VERSION 2. Update the Windows NVIDIA driver and WSL. Do not install a normal Linux NVIDIA display driver inside WSL.

5. If you are still stuck: send these diagnostics

Run these in Windows PowerShell:

```
wsl --version
```

```
wsl --status
```

```
wsl -l -v
```

Then run these inside Ubuntu:

```
uname -a
```

```
lsb_release -a
```

```
whoami
```

```
pwd
```

```
which python
```

```
python --version
```

```
which conda
```

```
conda --version
```

```
conda env list
```

Recommended project location

For class projects, work inside the Ubuntu filesystem rather than directly on the Windows Desktop:

```
mkdir -p ~/materials_by_design
```

```
cd ~/materials_by_design
```

To open that folder in Windows Explorer:

```
explorer.exe .
```